Personalized News Recommendation with Contextual Bandits Using NLP Embedding

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1. PROBLEM STATEMENT

There is an enormous amount of content available to users on modern web platforms. In order to show each user the most relevant content, news websites in particular have to filter thousands of pieces every day. Conventional recommendation systems frequently use ranking algorithms that are static and slow to adjust to user behavior. The goal of this project is to create a personalized news recommendation system that uses contextual bandit reinforcement learning algorithms to modify recommendations and natural language processing (NLP) embeddings to score news articles. First, the system will use text embeddings from article titles and descriptions to represent news articles. The model can calculate the semantic similarity between articles and user preferences thanks to these embeddings. A contextual bandit algorithm will dynamically modify recommendations depending on simulated user response on top of this ranking system.

2. RELATED WORK

News recommendations have been extensively studied in recent years. Neural approaches have shown strong performance by learning semantic representations of news content. Neural News Recommendation with Multi-Head Self-Attention introduced the NRMS model, which uses multi-head self-attention to learn representations of news articles and user interests from reading history. NPA: Neural News Recommendation with Personalized Attention proposed a personalized attention mechanism that captures user-specific preferences in news consumption. Deep Reinforcement Learning for List-wise Recommendations explored reinforcement learning approaches to recommendation systems, demonstrating improved adaptability to user interactions. Contextual bandits have also been widely applied in recommender systems. These methods balance exploration and exploitation while adapting to user feedback in real time. These works demonstrate the effectiveness of combining NLP-based content understanding with reinforcement learning approaches for adaptive recommendation systems.

3. SCOPE

The scope of this project includes: [1] News article text embedding generation, [2] Personalized ranking model, [3] Contextual bandit learning algorithm, [4] Offline simulation of user interactions, [5] Evaluation using ranking and reward-based metrics. The system will be evaluated using standard recommender system metrics such as nDCG and Hit@K, as well as reinforcement learning performance metrics such as cumulative reward. The system architecture for NLP and reinforcement learning integration has been partially implemented. In the NLP section, text preprocessing and generation of text embeddings using the TF-IDF and Sentence Transformer models are complete. These text embeddings are utilized to calculate the similarity score for the initial article ranking. In the reinforcement learning section, an epsilon greedy contextual bandit framework has been initialized, which enables the system to start learning from simulated user interactions.

4. DATASET

We made use of Misra's (2022) News Category Dataset, which we got from Kaggle, for this project. About 200,000 HuffPost news stories from 2012 to 2022 are available in 42 categories spanning over travel, wellness, politics, and entertainment. Every record has the headline brief description, category, author, and publication date. Since the dataset is made up only of published article metadata and contains no user-identifying information, it is publicly available on Kaggle for research purposes and poses no significant privacy issues. We selected this dataset because it accurately captures the variety of real-world news material that a personalized recommendation algorithm would likely come across.

5. BASELINE MODEL & RESULTS

The system architecture for integrating Natural Language Processing and Reinforcement Learning has been partially implemented. This includes a preprocessing step for the text of the news articles, generating embeddings for calculating similarity scores for the initial ranking. Additionally, a contextual bandit architecture has been initialized, enabling adaptive recommendations based on simulated user interactions. To set a

performance benchmark, two models have been implemented using the NLP technique. The first uses TF-IDF Vectorization coupled with Cosine Similarity for basic keyword matching. The second uses Sentence Transformer to establish semantic relationships between news articles and user preferences. These models are the basis of the improvements brought by the reinforcement learning technique. For the NLP aspect, various embedding models have been implemented and compared. A TF-IDF model is employed for the basic similarity of keywords, and SentenceTransformer is utilized for deeper semantic relationships between articles. These models are utilized for similarity score calculation for ranking purposes. Other transformer models, such as BERT, are being considered for further testing in future versions.

Figure 1: Data Pipeline Output

```
============================================================
STEP 1 — Data Pipeline
============================================================
[data_pipeline] Loaded 1,000 raw articles, 24 categories.
[data_pipeline] After filtering: 869 articles, 3 categories.
  Categories: ['ENTERTAINMENT', 'POLITICS', 'WORLD NEWS']
  Class distribution:
    POLITICS: 472
    ENTERTAINMENT: 259
    WORLD NEWS: 138
[data_pipeline] Split → train=695 | val=87 | test=87
[data_pipeline] After oversampling: 1,134 train samples ({'WORLD NEWS': 378, 'POLITICS': 378, 'ENTERTAINMENT': 378})
[data_pipeline] Saved splits to data\processed/
```

Figure 2: NLP Embeddings TF-IDF & SBERT Loading

```
============================================================
STEP 2 — NLP Embeddings
============================================================

[TF-IDF]
[TFIDFEmbedder] Fitted on 1,134 texts. Vocab size: 5,916
[TFIDFEmbedder] Saved to experiments\results\run3\tfidf.pkl
  Shape: (1134, 5916)  (0.1s)

[SBERT: all-MiniLM-L6-v2]
modules.json: 100%|##########################################| 349/349 [00:00<?, ?B/s]
D:\Files\School Code\Personalized-News-Recommendation-with-Contextual-Bandits-Using-NLP-Embedding\venv\Lib\site-packages\huggingface_hub\file_download.py:129: UserWarning: `huggingface_hub` cache-system uses s
ymlinks by default to efficiently store duplicated files but your machine does not support them in C:\Users\Jerel\.cache\huggingface\hub\models--sentence-transformers--all-MiniLM-L6-v2. Caching files will stil
l work but in a degraded version that might require more space on your disk. This warning can be disabled by setting the `HF_HUB_DISABLE_SYMLINKS_WARNING` environment variable. For more details, see https://hu
ggingface.co/docs/huggingface_hub/how-to-cache#limitations.
To support symlinks on Windows, you either need to activate Developer Mode or to run Python as an administrator. In order to activate developer mode, see this article: https://docs.microsoft.com/en-us/windows/
apps/get-started/enable-your-device-for-development
  warnings.warn(message)
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
config_sentence_transformers.json: 100%|####################| 116/116 [00:00<00.00, 11846/s]
README.md: 100%|#############################################| 10.5k/10.5k [00:00<?, ?B/s]
sentence_bert_config.json: 100%|#############################| 53.0/53.0 [00:00<?, ?B/s]
config.json: 100%|###########################################| 612/612 [00:00<?, ?B/s]
model.safetensors: 100%|#####################################| 90.9M/90.9M [00:03<00.00, 25.2MB/s]
Loading weights: 100%|#######################################| 103/103 [00:00<00.00, 8461.66it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

Figure 3: NLP Embeddings Saved Outputs

```
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%|#################################| 350/350 [00:00<?, ?B/s]
vocab.txt: 232kB [00:00, 4.86MB/s]
tokenizer.json: 466kB [00:00, 9.74MB/s]
special_tokens_map.json: 100%|###############################| 112/112 [00:00<?, ?B/s]
config.json: 100%|###########################################| 190/190 [00:00<?, ?B/s]
[SentenceEmbedder] Loaded 'all-MiniLM-L6-v2', embedding dim=384, device=cpu
Batches: 100%|#################################################| 6/6 [00:03<00:00,  1.59it/s]
[SentenceEmbedder] Saved embeddings (1308, 384) → experiments\results\run3\article_embeddings.npy
[SentenceEmbedder] Saved embeddings (1134, 384) → experiments\results\run3\train_embeddings.npy
[SentenceEmbedder] Saved embeddings (87, 384) → experiments\results\run3\test_embeddings.npy
```

Figure 4: SBERT and Logistic Regression

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============================================================
STEP 3c — SBERT + Logistic Regression (non-DL baseline)
============================================================
  Accuracy=0.8851  Macro-F1=0.8689
```

6. CNN EXPERIMENT

To supplement the Transformer-based word embeddings, we designed a high-performance 1D Convolutional Neural Network (1D-CNN) for headline classification and feature extraction. The model structure includes an Embedding layer followed by a 1D Convolutional layer with 128 filters and a kernel size of 5, enabling the model to learn local word correlation patterns (n-grams). Global Max Pooling was utilized to select the most significant features from the text input, which were then passed through a Dense layer. Initial results revealed that although the CNN model achieves faster convergence than the Sentence Transformer, the CNN fails to learn the deeper semantic nuances required for long-tail recommendation. These CNN features, however, offer a computationally efficient solution for the input state of the contextual bandit.

Figure 5: Embedding CNN Training Output

```
============================================================
STEP 3a — EmbeddingCNN (SBERT input) — PRIMARY CNN
============================================================
  Device: cpu  |  Classes: 3
  Epoch   1  loss=1.3038  val_acc=0.8966
  Epoch  10  loss=0.3139  val_acc=0.8851
  Early stopping at epoch 16.
[EmbeddingCNN] Done. Best val_acc=0.8966

[EmbeddingCNN Test]  Accuracy=0.8391  Macro-F1=0.8196
```

Figure 6: Text CNN from Scratch Training Output

```
============================================================
STEP 3b — TextCNN from scratch (ablation)
============================================================
[Vocabulary] Built vocab size=6,247
  Epoch  1  loss=3.6266  val_acc=0.4828
  Epoch  2  loss=2.4112  val_acc=0.5057
  Epoch  3  loss=1.7271  val_acc=0.5517
  Epoch  4  loss=1.3133  val_acc=0.6207
  Epoch  5  loss=1.1126  val_acc=0.6437
  Epoch  6  loss=0.7787  val_acc=0.6782
  Epoch  7  loss=0.7338  val_acc=0.6667
  Epoch  8  loss=0.6130  val_acc=0.6897
  Epoch  9  loss=0.4704  val_acc=0.6322
  Epoch 10  loss=0.4241  val_acc=0.7011
  Epoch 11  loss=0.3583  val_acc=0.6207
  Epoch 12  loss=0.2987  val_acc=0.6782
  Epoch 13  loss=0.2533  val_acc=0.7241
  Epoch 14  loss=0.2180  val_acc=0.7126
  Epoch 15  loss=0.2304  val_acc=0.7011
[TextCNN] Done. Best val_acc=0.7241

[TextCNN Scratch Test]  Accuracy=0.6437  Macro-F1=0.5833
```

Figure 7: CNN Training Loss & Validation Accuracy Curves

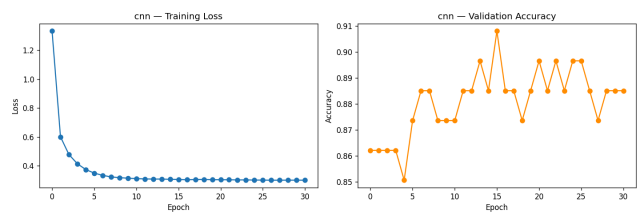
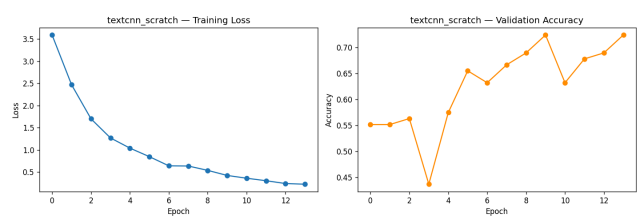


Figure 8: Text CNN Scratch Training Loss & Validation Accuracy



7. REINFORCEMENT LEARNING COMPONENT

The reinforcement learning part of the system uses the contextual bandit algorithm, which is based on the epsilon greedy algorithm. In this algorithm, the articles are selected based on a balance of exploration and exploitation, where $\varepsilon = 0.1$ controls the amount of exploration. In addition, the context of the articles is defined by the article embeddings from the NLP part of the system. A reward function is defined such that a click by a user gives a reward of +1, and no click gives a reward of 0. From the simulation results, the agent starts improving the quality of the articles over time, as measured by the cumulative reward.

Figure 9: RL Contextual Bandit Simulation

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============================================================
STEP 4 — RL Contextual Bandit Simulation
============================================================
  Random baseline: 0.0517
[rl_agent] Comparison plot saved → experiments\results\run3\rl_comparison.png
[rl_agent] Saved learning curve → experiments\results\run3\rl_epsilongreedy_curve.png
  EpsilonGreedy      mean=0.0600  total=180.0
[rl_agent] Saved learning curve → experiments\results\run3\rl_linucb_curve.png
  LinUCB             mean=0.0633  total=190.0
```

Figure 10: Agent Comparison, Rolling Average Reward

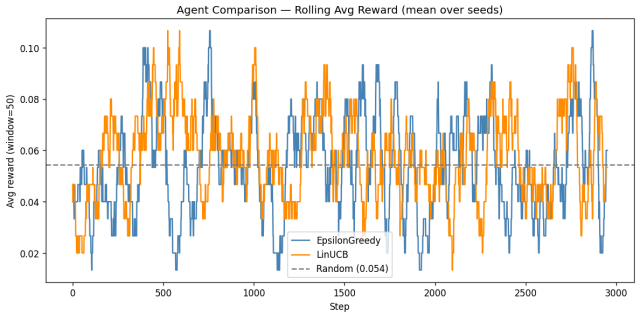


Figure 11: Epsilon Greedy Cumulative Reward & Rolling Average Reward

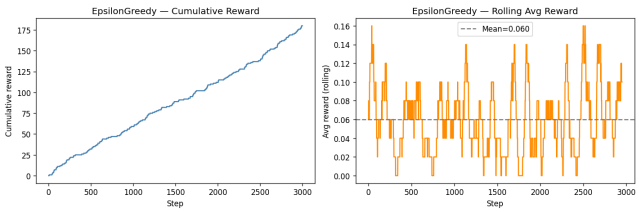


Figure 12: LinUCB Cumulative Reward & Rolling Average Reward

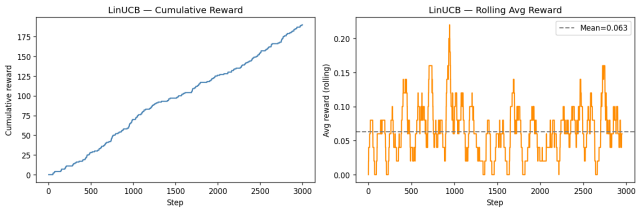


Figure 13: Confusion Matrix Normalized

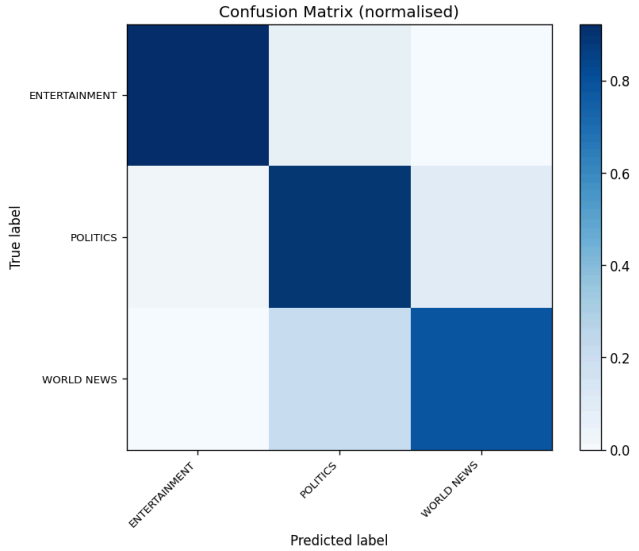
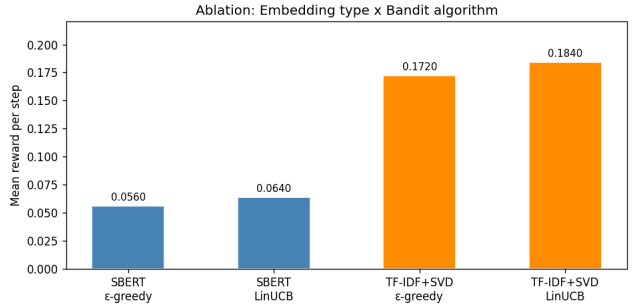


Figure 14: Ablation , Embedding Type × Bandit Algorithm



8. SUCCESS METRICS & CONSTRAINTS

The success will be measured using the following metrics: Ranking and Reinforcement Metrics. In Ranking Metrics the researchers will conduct Normalized Discounted Cumulative Gain (nDCG), and Hit Rate at K (Hit@K). In Reinforced Metrics the researchers will conduct Cumulative reward in offline simulation, and Click-through prediction accuracy. Constraints in the study will include: Limited computational resources, Offline evaluation due to lack of real added when the publications are assembled.

9. ETHICS STATEMENT

The development of a personalized news recommendation system using NLP embeddings and contextual bandit reinforcement learning raises several ethical considerations that have been addressed in the design of this project. First, personalized recommendation systems risk narrowing a user's exposure to diverse perspectives — because the contextual bandit agent optimizes for click-based rewards, it may reinforce existing user preferences and reduce content variety over time. To mitigate this, diversity constraints can be incorporated into the ranking logic to ensure that recommendations span multiple categories from the dataset's 42 news topics rather than converging on a narrow subset. Second, this project uses the publicly available News Category Dataset sourced from Kaggle, which contains only published article metadata with no user-identifying information, and relies on simulated rather than real user interactions throughout offline evaluation, which eliminates direct privacy risk. Any future extension involving real user behavior data should rely on anonymized interaction logs and adhere to applicable data protection standards. Third, since the dataset spans articles from 2012 to 2022, certain topics and writing styles may be overrepresented, and the reward function — which assigns +1 for a simulated click and 0 otherwise — may inadvertently favor high-visibility headlines over substantive content. Fairness-aware evaluation metrics and periodic rebalancing of training samples across categories are recommended to counteract this tendency. Finally, the use of Sentence Transformer embeddings and epsilon-greedy reinforcement learning introduces a degree of opacity into the recommendation process, and future iterations of the system should consider incorporating explainability mechanisms to make the basis of recommendations clearer to users and stakeholders.

10. TEAM ROLES & MILESTONES

To ensure a structured development process, the team is organized into specialized roles that cover each critical phase of the project:

- Member 1- Angel is responsible for coordinating the overall progress of the project. This includes organizing meetings, managing the project timeline, and ensuring that all required documents such as the proposal, reports, and presentations are completed on time. Angel also assists in organizing the dataset and documenting the system workflow to ensure that all components of the project are properly recorded.
- Member 2 - Gerail is responsible for collecting and preparing the dataset used in the project. This includes downloading the dataset, cleaning the data, organizing the files, and preparing the data so it can be used for training and testing the recommendation model. Gerail also assists in preparing the text data needed for generating embeddings.
- Member 3 – Jero focuses on implementing the Natural Language Processing (NLP) component of the system. This includes generating text embeddings from news articles and preparing the processed text data for use in the recommendation model. Jero also assists in integrating the embeddings into the recommendation pipeline.
- Member 4 – Kian is responsible for implementing the recommendation logic of the system. This includes building the ranking system and integrating the contextual bandit algorithm to improve recommendations based on simulated user interactions. Kian also performs the evaluation of the system by computing metrics such as nDCG and Hit@K and generating graphs or visualizations to explain the performance of the recommendation system.

For the project milestones, the researchers followed a structured timeline to move from initial planning to the final presentation:

- Week 1: Formalize the project proposal and complete the GitHub repository setup.
- Week 2: Execute the dataset preparation and preprocessing workflows. Develop and test the core NLP embedding model. Establish a baseline recommendation model for performance comparison. Complete the contextual bandit implementation for adaptive learning. Conduct final integration of all system components. Finalize the project report and deliver the concluding presentation.

11. REFERENCES

[1] Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. 2010. A contextual-bandit approach to personalized news article recommendation. *Presented at the Nineteenth International Conference on World Wide Web (WWW 2010), Raleigh, NC, USA, 2010* (April 2010), 661–670. https://doi.org/10.1145/1772690.1772758

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